

Assignment No - 06 (Group A)

Problem statement:

Data Analytics III

1. Implement Simple Naïve Bayes classification algorithm using Python/R on iris.csv dataset.
2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset

Pre-requisite

1. Basic of Python Programming
2. Concept of Joint and Marginal Probability.

Objective

Students should be able to data analysis using Naive Bayes Algorithm using Python for any open source dataset

Software and Hardware requirements:-

1. **Operating system:** Linux- Ubuntu 16.04 to 17.10, or Windows 7 to 10,
2. **RAM-** 2GB RAM (4GB preferable)
3. **IDE :-**Anaconda Jupiter Notebook / pycharm /Visual Studio

Theory-

1. Concepts used in Naïve Bayes classifier

Naïve Bayes Classifier can be used for Classification of categorical data.

Let there be a 'j' number of classes. $C = \{1, 2, \dots, j\}$

Let, input observation is specified by 'P' features. Therefore input observation x is given , $x = \{F_1, F_2, \dots, F_p\}$

The Naïve Bayes classifier depends on Bayes' rule from probability theory.

Prior probabilities:

Probabilities which are calculated for some event based on no other information are called Prior probabilities.

For example, $P(A)$, $P(B)$, $P(C)$ are prior probabilities because while calculating $P(A)$, occurrences of event B or C are not concerned i.e. no information about occurrence of any other event is used.

Conditional Probabilities:

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} \quad \text{if } P(B) \neq 0 \quad \dots \dots (1)$$

$$P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)} \quad \dots \dots (2)$$

From equation (1) and (2) ,

$$P(A \cap B) = P\left(\frac{A}{B}\right) \cdot P(B) = P\left(\frac{B}{A}\right) \cdot P(A)$$

$$\therefore P\left(\frac{A}{B}\right) = \frac{P\left(\frac{B}{A}\right) \cdot P(A)}{P(B)}$$

Is called the Bayes Rule.

2. Example of Naive Bayes

We have a dataset with some features Outlook, Temp, Humidity, and Windy, and the target here is to predict whether a person or team will play tennis or not.

Outlook	Temp	Humidity	Windy	Play
sunny	hot	high	FALSE	no
sunny	hot	high	TRUE	no
overcast	hot	high	FALSE	yes
rainy	mild	high	FALSE	yes
rainy	cool	normal	FALSE	yes
rainy	cool	normal	TRUE	no
overcast	cool	normal	TRUE	yes
sunny	mild	high	FALSE	no
sunny	cool	normal	FALSE	yes
rainy	mild	normal	FALSE	yes
sunny	mild	normal	TRUE	yes
overcast	mild	high	TRUE	yes
overcast	hot	normal	FALSE	yes
rainy	mild	high	TRUE	no

$$\underline{X} = [\text{Outlook, Temp, Humidity, Windy}]$$

$\underbrace{\hspace{1cm}}_{x_1} \quad \underbrace{\hspace{1cm}}_{x_2} \quad \underbrace{\hspace{1cm}}_{x_3} \quad \underbrace{\hspace{1cm}}_{x_4}$

$$C_k = [\text{Yes, No}]$$

$\underbrace{\hspace{1cm}}_{C_1} \quad \underbrace{\hspace{1cm}}_{C_2}$

Conditional Probability

Here, we are predicting the probability of class1 and class2 based on the given condition. If I try to write the same formula in terms of classes and features, we will get the following equation.

$$P(C_k | X) = \frac{P(X | C_k) \cdot P(C_k)}{P(X)}$$

Now we have two classes and four features, so if we write this formula for class C1, it will be something like this.

$$P(C_1 | x_1 \cap x_2 \cap x_3 \cap x_4) = \frac{P(x_1 \cap x_2 \cap x_3 \cap x_4 | C_1) * P(C_1)}{P(x_1 \cap x_2 \cap x_3 \cap x_4)}$$

Here, we replaced C_k with C_1 and X with the intersection of X_1, X_2, X_3, X_4 . You might have a question, It's because we are taking the situation when all these features are present at the same time.

The Naive Bayes algorithm assumes that all the features are independent of each other or in other words all the features are unrelated. With that assumption, we can further simplify the above formula and write it in this form.

$$P(C_1 | x_1 \cap x_2 \cap x_3 \cap x_4) = \frac{P(x_1 | C_1) * P(x_2 | C_1) * P(x_3 | C_1) * P(x_4 | C_1) * P(C_1)}{P(x_1) * P(x_2) * P(x_3) * P(x_4)}$$

This is the final equation of the Naive Bayes and we have to calculate the probability of both C_1 and C_2 . For this particular example

Outlook	Temp	Humidity	Windy	Play
Rainy	Cool	High	True	?

$$P(Yes | X) = P(Rainy | Yes) \times P(Cool | Yes) \times P(High | Yes) \times P(True | Yes) \times P(Yes)$$

$$P(Yes | X) = 2/9 \times 3/9 \times 3/9 \times 3/9 \times 9/14 = 0.00529 \rightarrow 0.2 = \frac{0.00529}{0.02057 + 0.00529}$$

$$P(No | X) = P(Rainy | No) \times P(Cool | No) \times P(High | No) \times P(True | No) \times P(No)$$

$$P(No | X) = 3/5 \times 1/5 \times 4/5 \times 3/5 \times 5/14 = 0.02057 \rightarrow 0.8 = \frac{0.02057}{0.02057 + 0.00529}$$

$P(No | Today) > P(Yes | Today)$ So, the prediction that golf would be played is 'No'.

Conclusion:

In this way we have done data analysis using Naive Bayes Algorithm for Iris dataset and evaluated the performance of the model